

My website



HY13dev.com

Ylya Martchenko

Full Stack Developer



About Me

Full Stack Developer with 3 years of corporate software engineering experience.

Finishing my advanced engineering degree, I am actively seeking a full-time position. Fully mobile and available to start on Monday, 13 July 2026.

Personal Information

Address:

9 en Chaplerue, F-57000 Metz

Phone number:

+33 6 99 72 35 79

E-mail address:

yl Yamartchenko@gmail.com

Driver's license & car

French: Native (C2)

English: Professionally Fluent (C1)

Russian: Mother Tongue (C1)

Skills

- Next.js, React, Angular
- Java (Spring Boot)
- HTML, CSS/SCSS
- JavaScript, TypeScript
- SQL, SQLite, PostgreSQL
- Git, Jira
- Node JS
- Python
- C, C++, C# (.NET)
- Docker, K8s

Education

> Engineer in IARN (Informatics, Automatic, Robotics, Networks)

2023 - 2026 | Polytech Nancy | Vandœuvre-lès-Nancy

> Diploma of Higher Education in Engineering Science

2022 - 2023 | F.R.U. Mathematics, computer science, mechanics | Metz

> BTEC Higher National Diploma in Physics Measures

2021 - 2022 | U.I.T. Department of Measures | Metz

> PeiP (Polytech engineering schools course)

2019 - 2021 | Polytech Nancy | Vandœuvre-lès-Nancy

Experience (More details on my website at the top)

✓ **2023/09** Equasens | Villers-lès-Nancy
(currently) **Full Stack Developer**

✓ **2025/06** Schumacher-Knepper | Wintrange, Luxembourg
2025/09 **Full Stack Developer**

HoodieYlya13 (Projects)

> Tesla Model 3 Mechatronics Integration (4th-Year Project)

Reverse-engineered a Tesla Model 3 Highland to design and manufacture a custom, 3D-printed turning signal stalk. Programmed an ESP32 microcontroller using asymmetric dual-core execution—allocating one core to safety-critical LIN-bus steering wheel data and the other to CAN-bus throughput—interfaced via Bluetooth Low Energy (BLE) with a custom mobile application built for real-time telemetry and diagnostics tracking.

> Next.js 16 Serverless RAG Engine & AI Digital Twin

Architected a zero-hallucination conversational AI utilizing a dual-layer RAG pipeline. Built a custom CI/CD automation flow that tokenizes GitHub repository manifests into high-dimension vector embeddings. Optimized the Next.js serverless architecture using Partial Prerendering (PPR) and Server-Sent Event (SSE) streaming to deliver real-time, context-aware codebase analytics.

> Multiplayer Real-Time Social Deduction Coding Game

Reverse-engineered minified production bundles from the Wayback Machine to revive and modernize an "Among Us" style collaborative game designed as an end-of-sprint team icebreaker. Rebuilt the infrastructure using Next.js 16 and React 19, deploying an edge WebSocket server with PartyKit and multiplexed Yjs CRDTs to handle real-time multiplayer states and synchronized code editing for up to 20 players. Integrated Pyodide (WebAssembly) to compile and execute Python code and unit tests entirely within the user's browser, eliminating latency and server-side execution vulnerabilities.